

DataCamp

RECIPE SITE TRAFFIC PREDICTION

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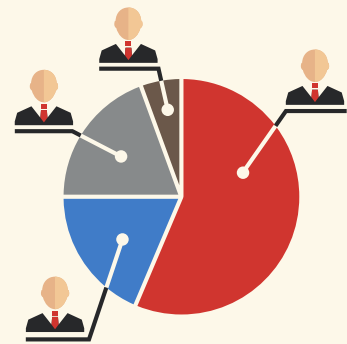
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Introduction



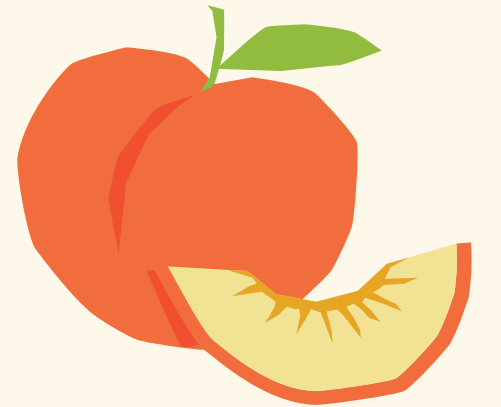
Working Process



Results



Recommendations



INTRODUCTION

Tasty Bytes is a recipe website that contain variety of menus.

As a data science team. We need to do a prediction of which recipes will be popular when displayed on the homepage website. They noticed that if they pick up popular recipes, the traffic to the website will grow up to 40%.



OBJECTIVES



Predict which recipes
will lead to high traffic?



Correctly predict high
traffic recipes 80% of
the time?

WORKING PROCESS



Data Validation
Checking the data
to fix the error and
fill missing values.



**Exploratory Data
Analysis**
Analyzing and
visualizing data.



Model Deployment
Making a model for
which recipe has high
or low traffic.



Model Evaluation
Compare the
performance
between 2 models.



Results
Make a summary and
recommendation.

DATASETS

947 Recipes



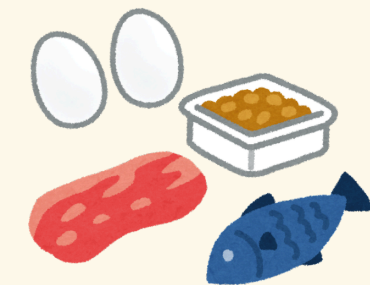
Calories
Number of calories



Carbohydrate
Amount of carbohydrates
in grams



Sugar
Amount of sugar in
grams



Protein
Amount of protein in
grams



Category
Type of recipe (Beverages,
Chicken, Meat, etc.)



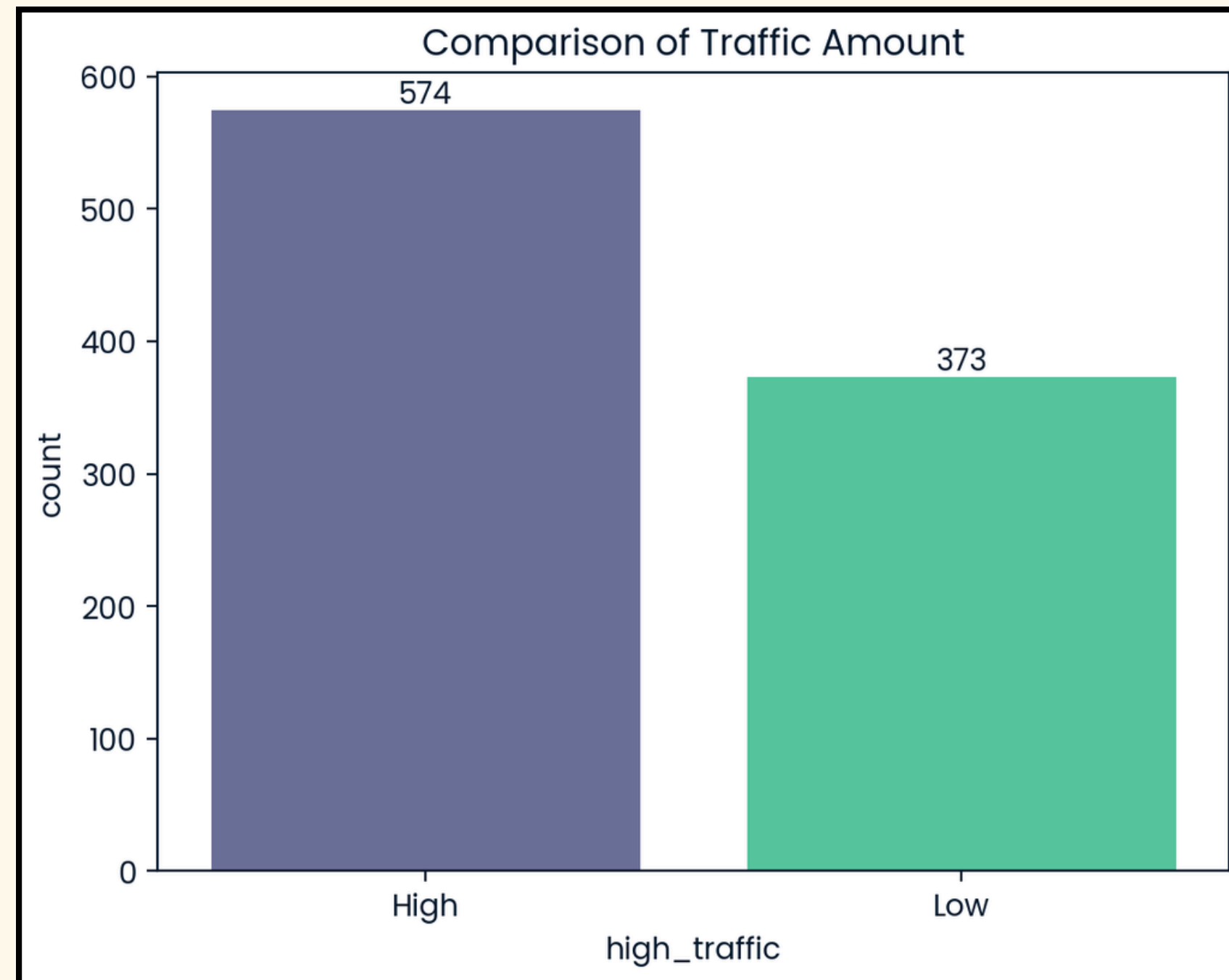
Servings
Number of servings
for the recipe



High Traffic
To show which recipe has
high traffic

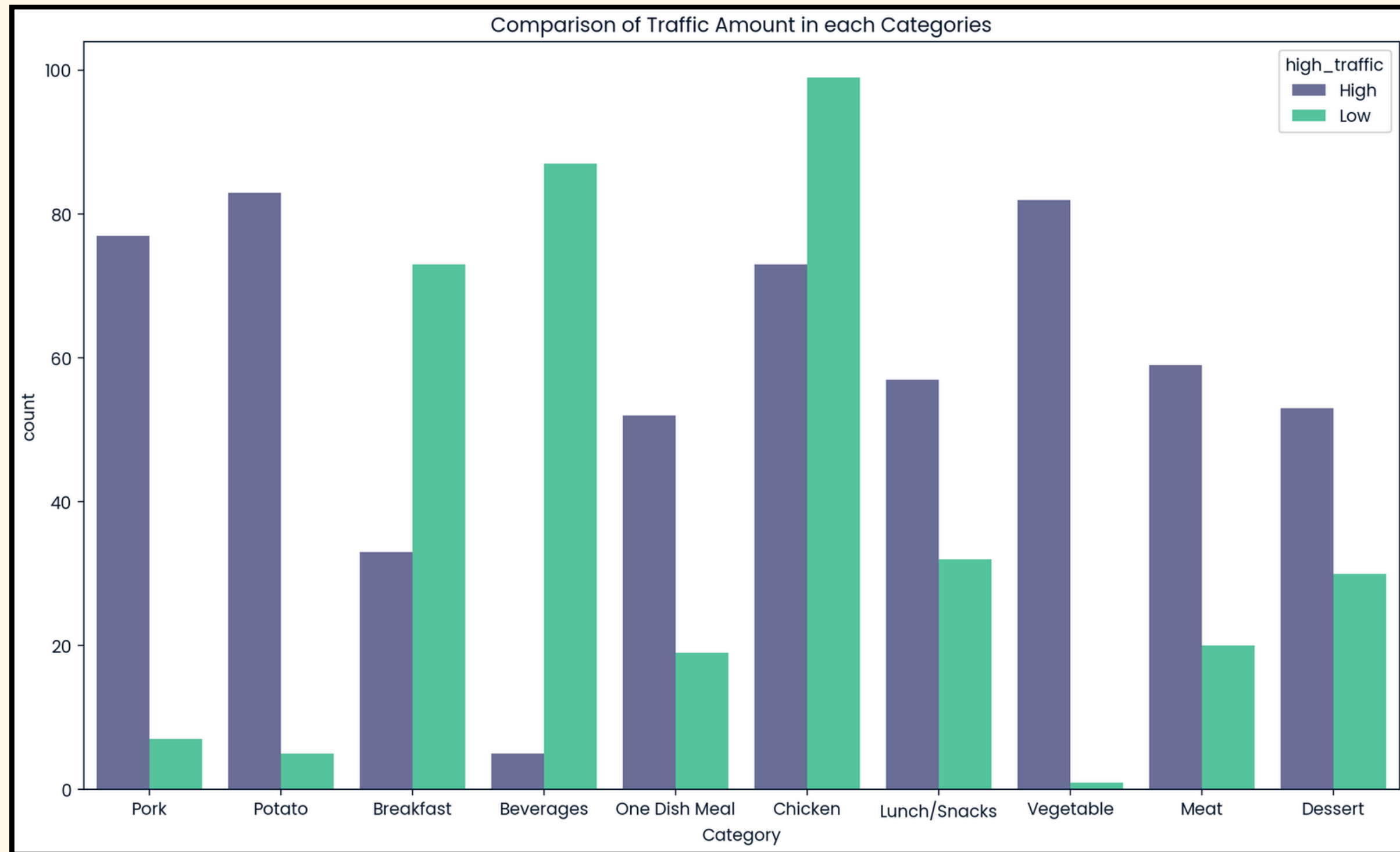


AMOUNT OF HIGH & LOW TRAFFIC





AMOUNT OF HIGH & LOW TRAFFIC WITH CATEGORIES





MODEL DEPLOYMENT PROCESS



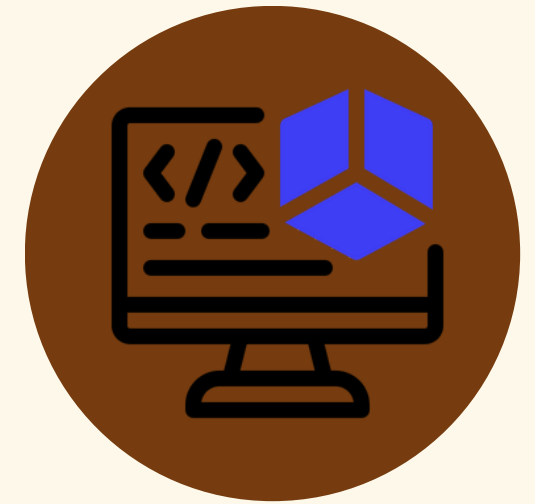
Feature Engineering
Encode feature to improve
model performance



Split features and label



Split training set and
test set to 80:20 ratio



Deploying model



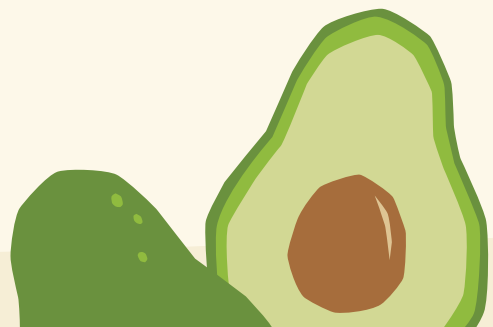
MACHINE LEARNING MODEL

Baseline Model

Logistic Regression is a supervised machine learning algorithm used for classification tasks in which the goal is to predict whether an instance belongs to a specific class or not.

Comparison Model

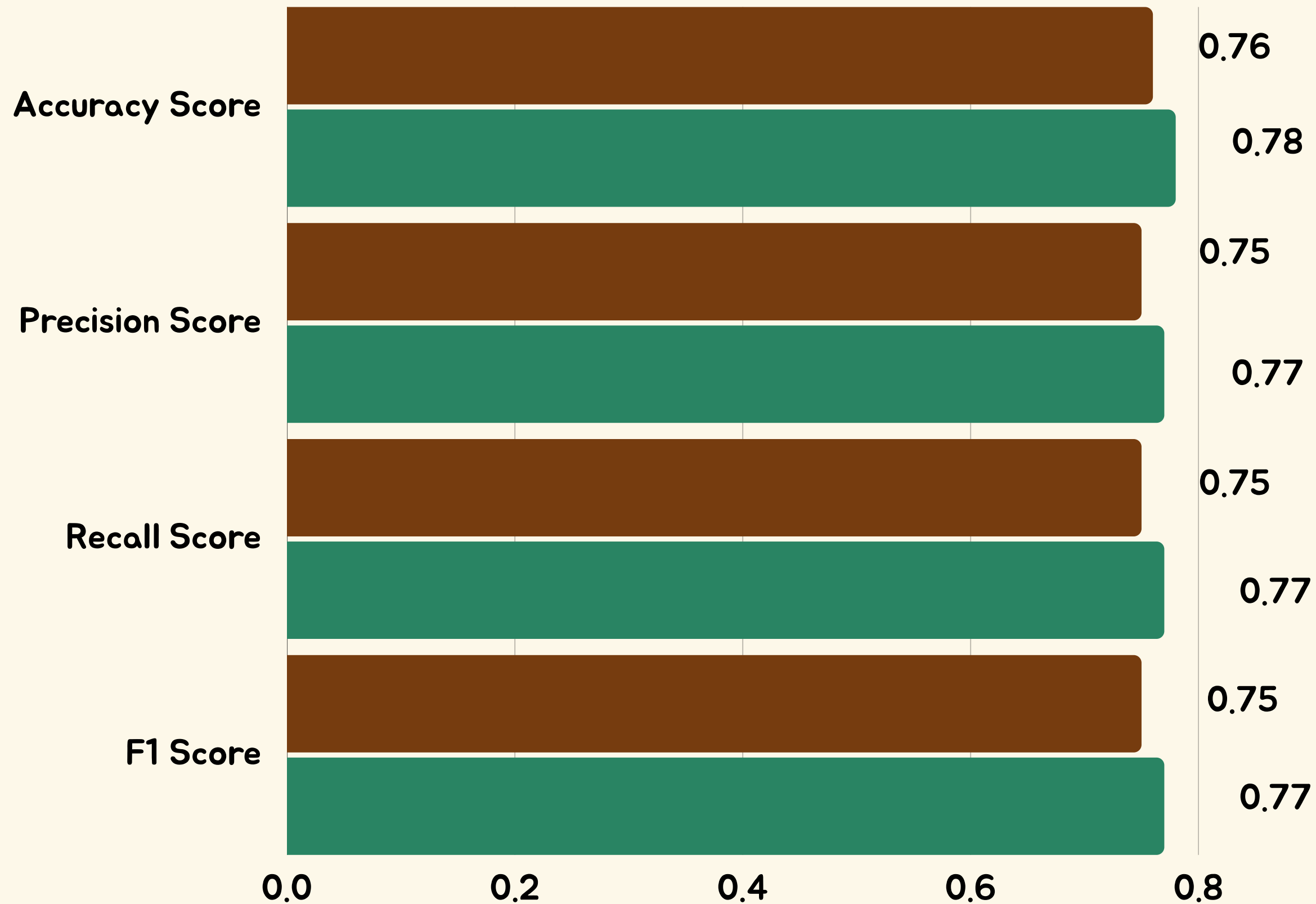
Random forests are a popular supervised machine learning technique that can handle both regression and classification tasks. Random forests are an ensemble method, which means they use predictions from multiple models.





MODEL EVALUATION

● Logistic Regression ● Random Forest



RECOMMENDATIONS



Deploying the Random Forest model as this model has better accuracy, precision, recall, and f1 score than the Logistic Regression model. Also with the precision, recall, and f1 score of predicting the high traffic.



Increasing the website traffic by displaying recipe categories like Vegetable, Potato, and Pork, which are the top 3 high amounts of traffic.



Collect more data like views to see how many times each recipe was viewed, ingredients of each recipe to attract users to be interested in which ingredients are used for each recipe, and the cost of recipes to improve the model performance.





THANK YOU

